Bridging Neural and Behavioural Data with XAI in Predictive Neuromarketing

Department of Computer Science

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Problem:

- Behavioral data (reaction times, gaze patterns) provide insights but fail to capture subconscious neural processes.
- Existing models:
  - Rarely integrate neural and behavioral data.
  - Are static and cannot adapt to dynamic contexts.
- Explainable AI (XAI) methods are underutilized.
- Ethical risks (bias, manipulation) remain largely unaddressed.

Impact/Gravity:

- Misunderstanding consumer behavior can lead to ineffective marketing strategies, poor user experience, and ethical concerns.
- Lack of transparency can reduce consumer trust in AI-driven recommendations.

Beneficiaries:

- **Marketers & Businesses:** More accurate, adaptable insights to optimize campaigns.
- **Research & Institutions:** Framework sets a new standard for ethical, multimodal neuromarketing research.

Design Excellence of the Solution:

- **Integrated Framework:** Combines neural signals (EEG) + behavioral metrics (reaction time, gaze patterns).
- **Adaptive & Dynamic:** Uses reinforcement learning to adjust predictions in real-time.
- **Explainable:** Implements SHAP and counterfactual techniques for transparency.
- **Ethical:** Includes trust and manipulation risk indices to safeguard consumer welfare.
- **Outcome:** Accurate, adaptable, transparent, and ethically sound insights into consumer decisions.

IT22150820 | KAVINDYA T A S

Title of the Individual Component: Behavioral Dynamics In Decision Making Scenarios

Specialization: Data Science

Behavioral Dynamics In Decision Making Scenarios

Solution:

- Uses behavioral data (mouse movement, gaze, reaction time) from PsychoPy
- Predicts preference percentages for brand, price, and familiarity
- Fully software-based and non-invasive

User Requirements Addressed:

- Enables researchers and marketers to understand which neural or behavioral features drive decisions.
- Supports actionable insights by allowing exploration of realistic behavioral or neural changes.
- Provides ethical oversight to ensure explanations are interpretable and non-manipulative.

Behavioral Dynamics In Decision Making Scenarios

Ethics & Transparency

- Uses only **observable user behavior**

- Simple, ethical, and easy to interpret

Design Excellence:

- **Behavioral Interpretability:**
  Links gaze, mouse movement, and reaction time to consumer decisions.
- **Clear Preference Modeling:**
  Predicts the influence of **brand, price, and familiarity** in an understandable way.
- **Ethical & Transparent Design:**
  Uses only observable behavior, avoiding invasive or manipulative techniques.
- **Modular Research Framework:**
  Flexible pipeline for data collection, preprocessing, and prediction, suitable for academic research.

IT22120984 | SENANAYAKA I H D M

Title of the Individual Component: Ethically Transparent XAI Framework for Consumer Decision Analysis

Specialization: Data Science

Ethically Transparent XAI Framework for Consumer Decision Analysis

Solution:

- Implements **dual-layer explainability** for neural and behavioral data.
- Provides **counterfactual scenarios** to explore “what-if” adjustments in user behavior or neural signals.
- Incorporates an **ethical scoreboard** to quantify transparency, bias, and manipulation risks.

User Requirements Addressed:

- Enables researchers and marketers to understand which neural or behavioral features drive decisions.
- Supports actionable insights by allowing exploration of realistic behavioral or neural changes.
- Provides ethical oversight to ensure explanations are interpretable and non-manipulative.

Ethically Transparent XAI Framework for Consumer Decision Analysis

Design Excellence:

- **Dual-Layer Explainability**: combines neural attribution and behavioral mapping.
- **Counterfactual Generation**: ensures actionable yet realistic scenarios.
- **Ethical Scoreboard Metrics**: Trust Index, Bias Detection, and Manipulation Risk Assessment embedded directly in insights.
- Modular, research-focused framework allowing transparent and interpretable analysis of consumer choice mechanisms.

IT22071316 | SHAHAAM M M

Title of the Individual Component: Contextual Decision Modelling & Adaptive AI using Federated Learning

Specialization: Data Science

Contextual Decision Modelling & Adaptive AI using Federated Learning

Solution:

- A **privacy-preserving predictive** neuromarketing system using **Federated Learning**
- Distributed client-side AI models simulate individual user contexts.
- A central federated server aggregates model intelligence without accessing raw data.
- Explainable AI (XAI) provides transparent reasoning behind predictions
- Real-time interaction through a React-based user interface

User Requirements Addressed:

- **Privacy & Data Protection**: User behavioral data remains on-device, Only model parameters are shared with the server
- **Context-Aware Decision Making**: Predictions adapt based on contextual inputs such as attention, dwell time, and device type.
- **Transparency & Trust**: XAI explanations show feature-level influence on decisions, Users can understand why a prediction was made

Design Excellence:

- **Modular Architecture** : Independent federated clients, Decoupled front end and back end, Central aggregation server

- **Scalability**: Easily extendable to more clients, Lightweight communication using REST APIs.

- **Ethical AI by Design**: Privacy-preserving learning approach, Explainable outputs reduce black-box risks.

- React for front-end interaction
- Flask for backend services
- Docker for consistent deployment and integration.

Commercialization & Market Potential

Customer Persona:

- Marketers & brands looking for better consumer insights.
- Research institutions exploring ethical neuromarketing.

Market Size:

- Growing global neuromarketing & AI analytics market.
- Opportunity in retail, e-commerce, and consumer research sectors.

Unique Features of Solution:

- Combines neural + behavioral data in real-time.
- Explainable & ethical insights.

Cost Recovery:

- Subscription-based SaaS for businesses.
- Licensing to research institutions

DEMO VIDEO